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QUARTERLY EXAMINATION - SEPTEMBER 2025

12TH STD CHEMISTRY

Time : 3: 00 HRS Maximum Marks : 70

PART I

Answer ALL the question. Choose and write the correct answer 15 x 1 =15

1. The metal oxide which cannot be reduced to metal by carbon is
   a) PbO b) Al_2O_3 c) ZnO d) FeO
2. Bauxite has the composition
   a) Al_2O_3 b) $Al_2O_3.H_2O$ c) $Fe_2O_3\ 3H_2O$ d)None of these
3. Which of the following statements is not correct?
   a) Beryl is a cyclic silicate b) Mg_2SiO_4 is an orthosilicate
   c) $[SiO_4]^{4-}$ is the basic structural unit of silicates d) Feldspar is not aluminosilicate
4. Treatment of gaseous boron trifluoride with sodium hydride around 450 K gives
   a) borane b) diborane c) boric acid d)None of these
5. Assertion : bond dissociation energy of fluorine is greater than chlorine gas
   Reason: chlorine has more electronic repulsion than fluorine
   a) Both assertion and reason are true and reason is the correct explanation of assertion.
   b) Both assertion and reason are true but reason is not the correct explanation of assertion.
   c) Assertion is true but reason is false. d) Both assertion and reason are false.
6. The actinoid elements which show the highest oxidation state of +7 are
   a) Np, Pu ,Am b) U, Fm, Th c) U, Th, Md d) Es, No, Lr
7. The radius of an atom is 300pm, if it crystallizes in a face centered cubic lattice, the length of the edge of the unit cell is a) 488.5pm b) 848.5pm c) 884.5pm d) 484.5pm
8. The crystal with a metal excess defect is
   a) NaCl b) FeO c) ZnO d) AgBr
9. The decomposition of phosphine (PH_3) on tungsten at low pressure is a first order reaction. It is because the
   a) rate is proportional to the surface coverage
   b) rate is inversely proportional to the surface coverage
   c) rate is independent of the surface coverage d) rate of decomposition is slow
10. Which of these is not likely to act as Lewis base?
    a) BF_3 b) PF_3 c) CO d) F^-
11. HPO_4^{2-} the conjugate base of
    a) PO_4^{3-} b) P_2O_5 c) H_3PO_4 d) HPO_4^{2-}
12. Isopropyl benzene on air oxidation in the presence of dilute acid gives
    a) C_6H_5-OH b) C_6H_5-CO-CH_3 c) C_6H_5-CO-C_6H_5 d) C_6H_5 - COOH
13. Phenol on catalytic hydrogenation gives cyclohexanol, the reaction is
    a) oxidation reaction b) nitrosation reaction c) sulphonation reaction d) reduction reaction
14. The formation of cyanohydrin from acetone is an example of
    a) nucleophilic substitution b) electrophilic substitution c) electrophilic addition d) Nucleophilic addition
15. Which one of the following reduces tollens reagent
    a) acetophenone b) acetic acid c) benzophenone d) none of these

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PART II

Answer any six questions. Question number 24 is compulsory 6 × 2 = 12

16. Give the basic requirement for vapour phase refining.
17. Give the structure of CO and CO_2.
18. What is inert pair effect?
19. Write the electronic configuration of Ce^{4+} and Co^{2+}.
20. Why ionic crystals are hard and brittle?
21. Give two examples for zero order reaction.
22. Write the expression for the solubility product of $Ca_3(PO_4)_2$
23. How will you prepare acetamide from methyl cyanide?
24. Classify the following alcohols as 1^0, 2^0, and 3^0 and give their IUPAC Names.
    a. $(C_2H_5)_3COH$ b. $CH_2=C(Cl)-CH(OH)CH_3$

PART III

Answer any six questions. Question number 33 is compulsory 6 × 3 = 18

25. How will you convert boric acid to boron nitride?
26. Write the reason for the anomalous behaviour of Nitrogen.
27. Write short notes on chromyl chloride test.
28. Distinguish tetrahedral and octahedral voids.
29. What are the differences between rate and rate constant?
30. Distinguish Lewis acid and Lewis base?
31. How will you convert acetylene into n-butyl alcohol?
32. Write short notes on Hell – Volhard – Zelinsky (HVZ) reaction.
33. How will you confirm that carbon can be used as a reducing agent above 1000 K to reduce FeO in the Ellingham diagram?

PART IV

Answer all the questions 5 × 5 = 25

34a. (i) Describe the role of the sodium cyanide in froth floatation. (2)
(ii) What are the differences between minerals and ores? (3)

(Or)

b. (i) What is the reaction of diborane with ammonia? (3)
(ii) Give the uses of silicones. (2)

35a. (i) What is the hybridisation of iodine in IF_7? Give its structure. (3)
(ii) Transition metals show high melting points. Why? (2)

(Or)

b. (i) Compare lanthanoids and actinoids. (3)
(ii) Classify the following solids a. P_4 b. Brass c. diamond d. NaCl. (2)

36a. (i) What is a body centered cubic (BCC) unit cell? Calculate the number of atoms in it. (3)
(ii) Define half-life of a reaction. (2)

(Or)

b. Derive an expression for Ostwald's dilution law. (5)

37a. (i) Find the half-life for the zero-order reaction. (3)
(ii) Give the uses of glycerol? (2)

(Or)

b. How will you differentiate 1^0, 2^0 & 3^0 alcohols using the Victor Meyer test? (5)

38a. (i) Why is the C-O-C bond angle in ether slightly greater than the tetrahedral bond angle? (2)
(ii) How will you prepare Cinnamic acid from benzaldehyde? (3)

(Or)

b. Write the mechanism of Aldol condensation reaction. (5)

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